

CyberVision AI — Project Execution Guide

Comprehensive Step-by-Step Setup, Configuration, and Troubleshooting Guide

1. System Architecture Overview

CyberVision AI is an enterprise threat intelligence and vulnerability management platform. The architecture is composed of two primary sub-systems working in tandem:

- **Backend API Server:** Powered by FastAPI (Python), APScheduler (background ingesters), NVD CVE API, RSS Scrapers, and Firebase Admin SDK.
- **Frontend Dashboard:** Next.js 16 (React 19, TypeScript, Tailwind CSS, Chart.js) providing a real-time SOC dashboard.

2. Prerequisites & Environment Setup

- **Python 3.10+** (with virtualenv enabled)
- **Node.js v18+ or v20+** (with npm 9+)
- **Firebase Credentials JSON:** Located at root or specified in backend .env file
- **NVD API Key:** Configured for automated threat intelligence scraping

3. Backend Setup & Startup Instructions

Step 3.1: Navigate to Backend Directory

Open a terminal and navigate to the backend folder:

```
cd backend
```

Step 3.2: Create and Activate Virtual Environment

On Windows (PowerShell):

```
python -m venv venv .\venv\Scripts\Activate.ps1
```

On Linux / macOS:

```
python3 -m venv venv source venv/bin/activate
```

Step 3.3: Install Dependencies

Install required Python packages:

```
pip install -r requirements.txt
```

Step 3.4: Configure Environment Variables (.env)

Ensure backend/.env contains valid paths and keys:

```
ENV=development
DEBUG=True
FIREBASE_PROJECT_ID=cybervisionai-35391
FIREBASE_CREDENTIALS_PATH=d:/cybervision/cybervisionai-35391-firebase-adminsdk-fbsvc-8862f9826d.json
NVD_API_KEY=ffe133eb-5c4d-4039-b5aa-338912a17059
```

Step 3.5: Launch FastAPI Backend Server

Run the server with Uvicorn (hot-reloading enabled):

```
python main.py # Or directly: uvicorn main:app --host 0.0.0.0 --port 8000 --reload
```

4. Frontend Setup & Startup Instructions

Step 4.1: Open Terminal in Frontend Directory

```
cd frontend
```

Step 4.2: Install Dependencies

```
npm install
```

Step 4.3: Launch Next.js Development Server

```
npm run dev
```

Step 4.4: Access the Frontend Dashboard

Open your web browser and navigate to: **http://localhost:3000**

5. Verification & Service Endpoints

Service	URL / Location	Expected Output / Purpose
Frontend UI	http://localhost:3000	CyberVision SOC Dashboard Interface
Backend Root API	http://localhost:8000/	{"status": "online", "database": "connected"}
Swagger API Docs	http://localhost:8000/docs	Interactive API specification & endpoints
ReDoc API Docs	http://localhost:8000/redoc	Alternative OpenAPI documentation
Backend Test Suite	cd backend && pytest	Runs pytest suite for repositories/scrapers

6. Troubleshooting Guide

- **Firebase Credential File Not Found:** Verify ``FIREBASE_CREDENTIALS_PATH`` in ``backend/.env`` points to the absolute path of your JSON key file.
- **Port Conflict (8000 or 3000 in use):** Identify and terminate conflicting processes, or specify ``--port 8001`` for uvicorn.
- **Missing Dependencies:** Ensure your virtual environment is active before running ``pip install -r requirements.txt``.
- **CORS Errors:** Ensure backend server is running and ``API_BASE_URL`` in ``frontend/utils/api.ts`` is configured to ``http://localhost:8000/api``.